
**SMALL UNMANNED AIRCRAFT SYSTEM OPERATOR
TRAINING PROGRAM**

JUNE 2026

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Small Unmanned Aircraft System Operator Training Program

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Preface

To effectively integrate unmanned aircraft systems across the force, the Army has established a proponent and executive agent structure. The Aviation Center of Excellence (AVCOE) serves as the single proponent responsible for establishing universal standards for unmanned aircraft system operation, airspace integration, and operator qualifications. The branch-specific centers of excellence (such as Maneuver, Fires, and Sustainment) act as executive agents and are responsible for developing the mission-specific tactics and training needed to employ UASs within their unique branch specific functions. This model ensures that while AVCOE provides guidelines to produce a licensed and mission certified operator; the executive agents determine how best to integrate that operator with their capabilities to achieve tactical effects.

TC 3-04.1-2 standardizes the Groups 1 and 2 small unmanned aircraft systems (SUASs) operator training program, operator licensing procedures, mission certification training, and SUAS gunnery across all centers of excellence (COEs). This training circular provides guidelines and requirements for COEs to develop mission training guidelines that are tailored to their specific needs. It is based on training principles outlined in ADP 7-0. This manual is not a stand-alone document; all the requirements contained in AR 95-1 must be met.

The principal audience for TC 3-04.1-2 is commanders, Master Trainers, and operators across all Army Branches. Commanders and staff of Army Headquarters serving as joint task force or multinational headquarters should also refer to applicable joint or multinational doctrine concerning the range of military operations and joint or multinational forces. Trainers and educators throughout the Army will also use this publication for reference, training, and qualification as appropriate.

Commanders, staffs, and subordinates ensure that their decisions and actions comply with applicable United States, international, and in some cases host-nation laws and regulations. Commanders at all levels ensure that their Soldiers operate in accordance with the law of war and the rules of engagement. (See FM 6-27.)

TC 3-04.1-2 uses joint terms where applicable. Selected joint and Army terms and definitions appear in both the glossary and the text. Terms for which TC 3-04.1-2 is the proponent publication (the authority) are italicized in the text and are marked with an asterisk (*) in the glossary. Terms and definitions for which TC 3-04.1-2 is the proponent publication are boldfaced in the text. For other definitions shown in the text, the term is italicized, and the number of the proponent publication follows the definition.

TC 3-04.1-2 applies to the Active Army, Army National Guard/Army National Guard of the United States and United States Army Reserve unless otherwise stated.

The proponent of TC 3-04.1-2 is the United States Army Aviation Center of Excellence (AVCOE). The preparing agency is the Directorate of Training and Doctrine, AVCOE. Send comments and recommendations on DA Form 2028 (*Recommended Changes to Publications and Blank Forms*) to Director, DOTD, ATTN: ATZQ-TDD (TC 3-04.1-2), 2218 6th Avenue, Fort Rucker, Alabama 36362; or by e-mail to usarmy.rucker.avncoe.mbx.doctrine-branch@army.mil, or submit an electronic DA Form 2028.

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Introduction

Small unmanned aircraft systems (SUASs) are critical enablers across all warfighting functions. The purpose of this publication is to establish a standardized baseline for all SUAS individual training, licensing, mission certification, and gunnery to ensure interoperability and safety across the Army.

This training circular establishes the mandatory minimum standards an individual must meet to be considered a licensed SUAS operator. These standards, set by AVCOE, are not intended to create a proficient operator. Instead, they provide the foundational baseline for safe and standardized operation. It is the responsibility of the various COEs to build upon these minimums by developing the specialized training programs required to forge proficient operators who are capable of executing mission-specific tasks. Commanders, trainers, and operators must adhere to the requirements herein to maintain uniformity and discipline across the force.

SUMMARY OF CHANGES

TC 3-04.1-2 fundamentally redesigns the SUAS Operator Training Program, replacing the legacy TC 3-04.62. It modernizes the overall training concept to account for SUAS proliferation, advanced payloads, and weaponized platforms across all warfighting functions. Major changes include:

- Multi-tiered progression replaces previous training models with three distinct, sequential pathways: Initial License Training (Chapter 3), Mission Certification (Chapter 4), and Gunnery Qualification (Chapter 5).
- Standardized six-table framework implements a mandatory, progressive six-table structure (academics to live-flight evaluation) across all training pathways to ensure Army-wide standardization.
- Proponent baseline vs. unit execution establishes a universal flight-skills baseline for Initial Licensing Training, while intentionally granting commanders and COEs the flexibility to build unit-specific mission-essential task list (METL) tasks into the Mission Certification and Gunnery Qualification pathways.
- Similar series SUAS introduces a categorization framework (Figure 3-3) based on platform control methods to streamline training and reduce redundant qualification requirements for operators flying multiple systems.
- Modernized tactical requirements integrates updated standards for combat identification (CID), non-organic and advanced payload instruction, and lethal target prosecution.
- Updated administration: refines commander's evaluation, hands-on skills assessment, currency, and annual proficiency requirements to ensure sustained operational readiness.

WORD DISTINCTIONS

Throughout the manual, specific terminology or terms of reference are used with specific meaning. The key terms of use are described in the following sub-paragraphs.

For clarity and to prevent divergence between COEs, the following terms are standardized in this circular:

- Currency: The mandatory recurring flight requirements an operator must meet to maintain their Mission Certified status without incurring additional training or evaluation. Currency standards are prescribed in Chapter 2 and must be enforced uniformly across all units.
- Equipment procedures: this encompasses all actions required to prepare for and conclude a flight mission, as specified in the platform's technical manual. This includes, but is not limited to pre-flight assembly, forms and records inspection, system power-up, control and data link verification, pre-flight checks, post-flight shutdown, inspection, and operator-level maintenance actions.
- Fixed-wing SUAS: An aircraft that generates aerodynamic lift solely through the forward motion of the aircraft across stationary wings. Fixed-wing platforms cannot hover, require continuous forward

airspeed to maintain flight, and typically necessitate specific launch (for example hand-launch, equipment launched, catapult) and recovery procedures.

- **Gunnery Qualification Training:** The specialized, progressive six-table training pathway (detailed in Chapter 5) designed for Mission Certified operators. This program develops the technical and tactical proficiency required to accurately acquire, identify, and prosecute targets using weaponized or lethal-capable SUAS platforms.
- **Gunnery Qualified:** The formal designation and operational status awarded to an operator upon the successful completion of the culminating live-fire evaluation (Gunnery Table VI). This status validates that the operator has demonstrated proficiency to Army standards and is authorized to employ weaponized SUAS platforms and lethal effects during collective live-fire operations and combat.
- **Hands-on skills assessment (HSA):** A formal, evaluated flight conducted by a MT or UT to determine, verify, or re-establish an operator's flight status. An HSA assesses an operator's proficiency in aircraft control and adherence to safety standards, and it is typically administered following a lapse in currency, a safety mishap, or when initial records are unverifiable.
- **Licensing:** The foundational training pathway (Initial License Training Tables I-VI) and formal designation awarded to an operator who has demonstrated safe, proficient, and standardized flight operations of a specific SUAS platform.
- **Master Trainer (MT):** The unit's subject matter expert for SUAS training, evaluation, and standardization. The MT is a graduate of the SUAS MT Course, Instructor Operator Course, or MT equivalency training. The SUAS MT works with the commander to implement training procedures that license, certify, and qualify SUAS operators, maintain records, and oversee UTs.
- **Mission Certified:** The operational status awarded to an operator upon the successful completion of the formal evaluation flight (Mission Certification Table VI). It validates the operator's proficiency in tactically employing the SUAS in support of the unit's METL and authorizes them to participate in collective training and combat operations.
- **Mission preparation:** The training status and progressive learning phase (Mission Certification Tables I-V) where a licensed operator learns, practices, and integrates mission-level academics and tactical flight tasks prior to their final certification evaluation.
- **Re-designation:** The process of formally re-evaluating an operator's status due to a loss of currency, demonstrated deficiency, or failure to meet evaluation standards. The outcome of re-designation determines what, if any, re-training is required.
- **Re-training:** The overarching term for any directed training required to correct an identified performance deficiency. Re-training can range in scope from the immediate correction of a single task to re-designation.
- **Rotorcraft SUAS:** An aircraft that generates aerodynamic lift and propulsive thrust exclusively from one or more engine-driven rotors throughout its entire flight profile (for example quadcopters, traditional helicopters).
- **Unit Trainer (UT):** An individual trained and evaluated to assist the MT in conducting SUAS training. A UT has not graduated from an approved formal MT course and must conduct instructional training under the supervision of a certified MT.
- **Vertical takeoff and landing SUAS:** A hybrid aircraft capable of taking off, hovering, and landing vertically like a rotorcraft, but which transitions to fixed-wing aerodynamic lift during forward flight to achieve greater speed and endurance.

Will, shall, must, should, can, and may are utilized as follows:

- ***Will, shall, or must*** indicate a mandatory or required item that is not optional.
- ***Should*** indicates a preferred but not mandatory item, task, or method of accomplishment. These are recommended best practices for the unit to complete.
- ***Can* or *may*** indicate an acceptable option, level of flexibility, or consideration for the unit.

The following words emphasize important and critical instructions and are integrated into SUAS operator tasks, actions, processes, or procedures as required:

- ***Warning.*** A warning is an operating procedure or a practice which, if not correctly followed, could result in personal injury or loss of life.

- **Caution.** A caution is an operating procedure or a practice which, if not strictly observed, could result in damage to or destruction of equipment.
- **Note.** A note highlights essential information of a non-threatening nature.

UNMANNED AIRCRAFT SYSTEM GROUPS

This training circular only applies to SUAS groups 1-2 referenced in JP 3-30. The unmanned aircraft system categorization chart (Table Introduction-1) outlines the typical weights, altitude, and air speeds associated with groups 1-5 systems. The associated academics and tasks within this training circular are the required minimums for a SUAS operator to be considered licensed, mission certified, and gunnery qualified to conduct unit level missions.

Table Introduction-1. Unmanned aircraft systems categorization chart

Category	MGTW (lbs.)	Normal Operating Altitude (ft)	Speed (KIAS)
Group 1	0 - 20	< 1200 AGL	100
Group 2	21 - 55	< 3500 AGL	< 250
Group 3	< 1320	< 18,000 MSL	
Group 4	>1320		> 18,000 MSL
Group 5		Any Speed	
Legend: AGL: above ground level lbs.: pounds ft: feet MGTW: maximum gross takeoff weight KIAS: knots indicated airspeed MSL: mean sea level			

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Chapter 1

Roles and Responsibilities

The successful implementation of the small unmanned aircraft system (SUAS) Operator Training Program (OTP) depends on the strict adherence to the roles and responsibilities defined within this chapter. This section delineates the specific duties required to manage, execute, and evaluate the progressive training pathways (Initial Licensing, Mission Certification, and Gunnery Qualification). It establishes the chain of accountability from the commander's program oversight down to the individual operator's technical and tactical proficiency.

COMMANDER

1-1. Commanders are the unit's primary training managers in accordance with ADP 7-0. They establish and resource the SUAS OTP, enforce standardization, certify MTs and UTs, manage risk, and ensure compliance with this circular and governing regulations. Commanders approve licensing, mission certification, qualification, re-designations and adjudicate waivers as permitted by AR 95-1.

1-2. They are responsible for ensuring unit standard operating procedures (SOPs) prescribe the completion of minimum training of this training circular and any additional tasks they deem necessary for the unit's mission.

MASTER TRAINER

1-3. The Master Trainer (MT) is the commander's technical advisor. The MT assists the commander in development, implementation, and management of the OTP. The MT is a graduate of the SUAS MT Course, Instructor Operator Course, or MT equivalency training. The MT is designated to conduct training and evaluations of all SUAS operators and Unit Trainers. To perform MT duties, the MT will be licensed, mission certified, and current in the SUAS or a similar series SUAS in which the MT will be performing their duties. An MT will be appointed by the first field grade commander with organic SUAS assets in their formation. It is highly recommended that MTs be rated directly by the OTP commander as they are the direct advisors.

1-4. The MT will document an operator's training. They will ensure all required training is captured in the Soldier's individual training folder or approved electronic record and submitted to the Army's designated training management system (see AR 350-1).

1-5. The MT will document operators who fail to meet standards and provide re-training before that operator can resume operating SUAS equipment. SUAS operators who fail re-training should have their operator's status re-designated by the OTP commander.

1-6. MTs maintain operational readiness of all SUAS equipment. They track system status, manage components, and submit reports to the OTP commander.

UNIT TRAINER

1-7. The Unit Trainer is trained and evaluated to fill the roles and duties of an MT but has not graduated one of the approved MT courses described in paragraph 1-3. To perform UT duties, UTs will be licensed, mission certified, and current on the SUAS or a similar series SUAS in which they will be performing duties. UTs must be trained and evaluated by an MT and will be documented on the operator's individual training folder or approved electronic record and submitted to the Army's designated training management system. UTs assist MTs in their roles and duties after undergoing training and evaluation by the supervising MT. A UT will be appointed by their OTP commander and annotated in the appropriate training folder.

1-8. A designated UT is authorized to conduct Initial License Training under the supervision of an MT. However, the licensing evaluations must be conducted and recorded exclusively by an MT, except when a unit is utilizing the self-start provisions outlined in paragraphs 2-26 through 2-28.

1-9. UTs may not evaluate an MT except to reestablish currency.

SMALL UNMANNED AIRCRAFT SYSTEM OPERATOR

1-10. Operators perform system setup, mission planning, launch, applicable payload operations, and recovery. Operators will maintain the standards and currency per this training circular. They will maintain a working knowledge of prescribed subject requirements, mission specific tasks, and the skills necessary to safely operate all systems they are certified and approved to operate. They must be licensed and mission certified by the commander to operate SUAS platforms without MT or UT supervision.

Chapter 2

Operator Training Program

This chapter establishes the mandatory framework and directives for the unit-level Operator Training Program (OTP). The purpose is to equip commanders and Master Trainers (MTs) with a systematic approach to manage, certify, and sustain SUAS operators throughout their entire lifecycle. This framework governs the progression from the initial commander's evaluation of newly assigned personnel through the structured training pathways required to achieve SUAS Licensed, Mission Certified, and Gunnery Qualified designations. Furthermore, it details the evaluation tools, currency rules, and annual proficiency standards necessary to ensure all operators remain proficient in their mission-essential task list (METL), fully integrated into the unit's warfighting function, and maintained at the highest level of readiness.

COMMANDER'S EVALUATION

2-1. The commander's evaluation verifies SUAS operator license, determines platform certifications, and determines mission level certification of newly assigned operators. This evaluation consists of a records review and a hands-on skill assessment (HSA), if required. The commander or MT will complete the evaluation within 45 calendar days after the operator is assigned for duty as a SUAS operator.

RECORDS REVIEW

2-2. Unit commanders, or their designated representative (MT or UT), will review the Soldier's individual training folder or approved electronic record and verify entries in the Army's designated training management system. The commander and designated representative will compare the Soldier's licensing and platform certifications history with the tasks required by the assigned duty position. If the appropriate mission level status can be clearly determined from the records review, the commander may approve an operator as mission certified without requiring an HSA.

2-3. SUAS operators must also request a copy of their individual training folder or approved electronic record and confirm that entries in the Army's designated training management system are current upon receipt of permanent change of station orders. SUAS operators will provide a copy of their training record to their gaining unit's commander or the commander's representative within 14 days of arrival to the unit. This ensures that the gaining unit has records of previous SUAS training and allows the MT or UT to identify training deficiencies.

HANDS-ON SKILL ASSESSMENT

2-4. A hands-on skill assessment will be utilized in the following scenarios:

- If the initial mission-level status cannot be determined by a record review or if the commander desires, the operator will undergo an HSA. The results of the HSA determine the operator's mission-level status. The commander or designated representative documents the mission level status on the operator's individual training folder or approved electronic record and ensures entries are reflected in the Army's designated training management system.
- If a SUAS operator lapses in currency.
- If a training deficiency or any unsafe act occurs (for example an accident, mishap, unsafe landing, or airspace violation).

2-5. An HSA should be conducted utilizing the applicable methods of control and aligned with the unit's METL; additional tasks can be added by the evaluator or commander from areas in which the operator can expect to perform duties. Methods of control include:

- Manual, non-stabilized, human-in-the-loop
- Manual, stabilized, human-in-the-loop
- Semi-autonomous, human on the loop
- Autonomous, human out of the loop

Note. An HSA can be conducted in a simulated environment except when required to re-establish currency.

RE-TRAINING AND RE-DESIGNATION

2-6. An operator can be re-designated on a platform or mission-level status for a training deficiency at the discretion of the unit commander. The operator must be re-trained on all deficiencies or if they have not flown within the previous 180 days. The MT or UT is responsible for documenting the deficiencies and the subsequent re-training or re-designation.

Note. The commander may choose to remove or change any approval granted to an operator, at their discretion. This event is recorded in the operator's individual training folder or approved electronic record and entered in the Army's designated training management system. This removal may be temporary, permanent, or because the commander determined that additional training is required.

OPERATOR TRAINING TABLES

2-7. The operator training tables are the Army's progressive, standardized method for training, licensing, certifying, and gunnery qualifying SUAS operators. Administered by an MT or UT, this framework consists of three distinct training pathways:

- Initial License Training (Tables I-VI).
- Mission Certification (Tables I-VI).
- SUAS Gunnery Qualification (Tables I-VI).

INITIAL LICENSE TRAINING

2-8. Initial License Training establishes the Army's mandatory baseline requirements for all SUAS operators. Before an operator is authorized to begin mission specific training or conduct operational flights, they must first earn the designation of "Licensed". This license is achieved by successfully completing the six-table Initial License Training pathway detailed in Chapter 3.

MISSION CERTIFICATION

2-9. Once Licensed, an operator advances to Mission Certification Training (Chapter 4). The purpose of this training is to transition the operator from basic flight proficiency to tactical readiness by focusing on the specific tasks required to support their unit's METL. During this pathway, an operator enters a "Mission Preparation" status to learn and practice these tasks. Upon successful evaluation, the operator earns the "Mission Certified" designation and is authorized to conduct tactical missions.

2-10. Active Army SUAS operators complete Mission Certification Training by passing mission level academics, combat identification, and mission related flight tasks. Once an operator begins this training, they have 90 consecutive days to advance from mission preparation status to a mission certified status. Once Mission Certified, operators must perform mandatory currency and proficiency requirements to maintain this status. Army National Guard and U.S. Army Reserve operators have one year to complete this requirement.

GUNNERY QUALIFICATION

2-11. A Mission Certified operator whose duties require the employment of lethal capable SUAS platforms must achieve Gunnery Qualification. This distinct training pathway, detailed in Chapter 5, is designed to train and qualify an operator to accurately acquire, identify, and prosecute targets. Successful completion of the Gunnery tables results in the operator being designated Gunnery Qualified.

TRAINING PRINCIPLES

2-12. The integrity and value of the SUAS training program depend on strict adherence to the following fundamental principles:

- Sequential progression: within any given pathway (Initial License, Mission Certification, or Gunnery Qualification), all tables are designed to be completed sequentially. An operator must successfully complete all events in a table before advancing to the next.
- Authorized concurrency: To maximize training efficiency, commanders are authorized to conduct specific tables concurrently (Figure 2-1). However, the following prerequisites must be met:
 - Mission Certification: an operator may complete Mission Certification Tables I-II concurrently with Initial License Tables I-II. However, an operator must fully complete the Initial License Training pathway and be awarded an operator license before they can progress to Mission Certification Tables III through VI.
 - SUAS Gunnery: an operator may complete SUAS Gunnery Tables I-II concurrently with Initial License or Mission Certification tables. However, an operator must fully complete the Mission Certification pathway and be signed off as Mission Certified before they can progress to SUAS Gunnery Tables III through VI.
- Commander's discretion: while concurrency is authorized, commanders should carefully consider the potential for operator information overload, time constraints, and resource availability before directing simultaneous training events.
- Standardization: the requirements for each table in this circular are mandatory minimums. Units may add tasks, but they may not remove, reduce, or alter the existing baseline standards.
- Certified evaluation: all formal evaluations must be administered by an MT or UT.
- Safety: operators must demonstrate 100% compliance with safety procedures. Violations result in automatic failure and retraining.

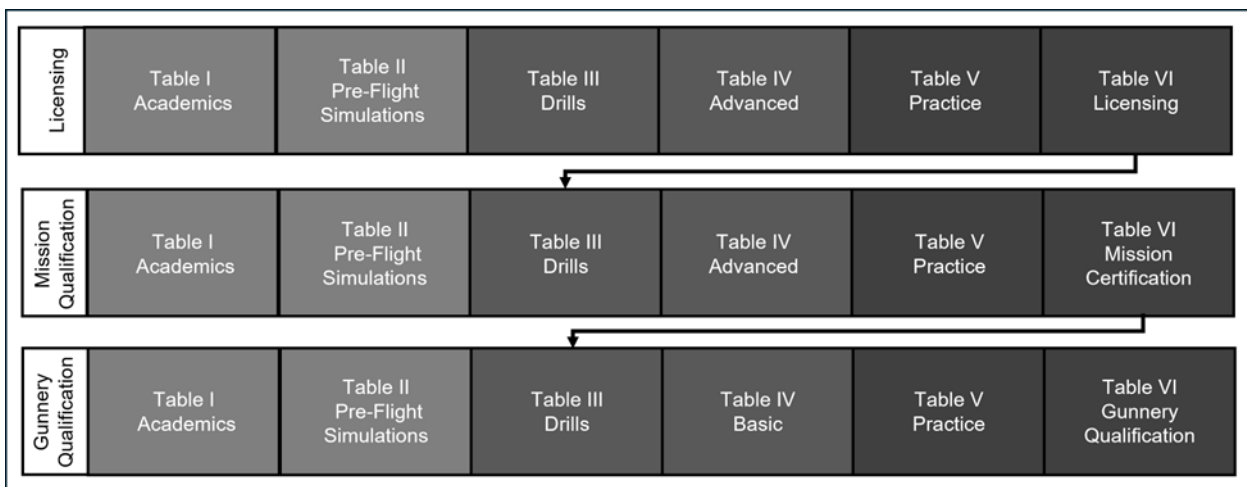


Figure 2-1. Concurrent academic flow chart

EXCLUSION PERIODS

2-13. An exclusion period is defined as time lost, at no fault of the operator, during Mission Certification training. This provision pauses the operator's allotted 90-day training timeline to prevent penalization for unavoidable delays. An authorized exclusion period consists of days lost due to:

- Temporary duty or deployment to a location where the operator is unable to fly.
- Medical or nonmedical suspension from operations.
- Leave approved by the unit commander.

2-14. If the exclusion period exceeds 45 consecutive days, the operator must restart the mission-level progression cycle, and the operator will have 90 consecutive days from this point to progress to a Mission Certified status. When an operator has not progressed within the required period, the unit commander will investigate and based on those findings, act according to AR 95-1.

SIMILAR SERIES

2-15. The proliferation of multiple manufacturers for SUAS has increased the number of systems that are being utilized by units across the force. While the systems have unique characteristics, generally systems share control and operation characteristics. Conducting Initial Licensing Training on each system is cumbersome and will reduce time available to conduct Mission Certification Training. Unit Master Trainers should utilize expertise to identify systems that share enough characteristics and qualities in order to reduce Initial Licensing Training time.

2-16. The following framework in Table 2-1 provides Commanders and MTs with a two-step process for identifying similar platforms. Step 1 is to identify what category the system planned for training falls into. Step 2 allows units to identify the type of control required for that system.

Table 2-1. Similar series small unmanned aircraft systems

Step 1: SUAS CATEGORIES	Step 2: METHOD OF CONTROL
ROTORCRAFT	MANUAL (Non-Stabilized [HITL])
VTOL (tilt rotor/lift and cruise)	MANUAL (Stabilized [HITL])
FIXED WING	Semi-Autonomous (HOTL)
OTHER SUAS	Autonomous (HOOTL)
Legend HITL: human in the loop HOTL: human on the loop HOOTL: human out of the loop	SUAS: small unmanned aircraft system VTOL: vertical takeoff and landing

Note. Unit Master Trainers recommend similar series SUAS platforms to the first O-6 in the chain of command for approval.

Note. Task iterations, currency, and evaluation requirements may be completed in similar aircraft.

2-17. Once an operator has successfully completed the Initial License Training pathway in its entirety on their primary system, they may utilize an abbreviated progression to earn a license on subsequent platforms. The specific training requirements for transitioning to a new SUAS series platform are determined as follows:

- Similar platform: an operator will complete an abbreviated Table I (Hands-on Platform Instruction only) followed by Tables III and IV to demonstrate live-flight proficiency before completing Table VI for the license. Table V may be waived at the commander's discretion.
- Dissimilar platform: an operator will complete an abbreviated Table I (Hands-on Platform Instruction only), followed by Table II (Simulations) and Tables III and IV to demonstrate live-flight proficiency before completing Table VI for the license. Table V may be waived at the commander's discretion.

Note. It is highly recommended to attempt to utilize one system throughout training.

Note. If the user interface hardware or software differ significantly and it makes basic operator control challenging to understand across systems, system specific re-training is recommended.

Note. The selection of a display system, either first-person view goggles or a personal electronic device, is a deliberate tactical decision with significant trade-offs. First-person view goggles provide superior flight immersion and are ideal for dynamic maneuvers or operating in high-glare environments, but they render the operator blind to their immediate surroundings. Conversely, a personal electronic device allows for greater situational awareness of the operator's local environment, but it is susceptible to screen glare. Commanders and operators must weigh these considerations based on a thorough analysis of mission variables prior to flight to ensure the benefits of the chosen system outweigh the risks for the specific mission.

CURRENCY REQUIREMENTS

2-18. To be considered current, an SUAS operator must, every 60 days, perform a takeoff, flight route, and landing through either live flight or an approved simulator. Every 90 consecutive days, they must perform a live takeoff, flight route, and landing on an SUAS they are certified on or a similar series. The SUAS operator whose currency has lapsed must complete a hands-on skill assessment according to paragraph 2-4 of this training circular. Simulators may not be used to reestablish currency.

Note. Currency only applies to SUAS operators that are in a Mission-Certified status.

Note. The "flight route" requirement is intended to ensure that the SUAS operator actively demonstrates safe operation and flight proficiency by utilizing the aircraft's equipment and maneuvering capabilities during flight.

Note. Takeoffs, flight routes, or landings performed while the SUAS is connected to a tether do not count toward currency requirements.

2-19. An SUAS operator having not flown an SUAS platform they are certified on, or similar series, within the previous 180 consecutive days will be re-designated to a mission preparation status and will receive re-training.

STANDARDIZATION PROGRAM

2-20. The commander is responsible for the unit's standardization program and must integrate it throughout the overall training strategy. Standardization must be implemented in all training tasks and is primarily overseen by MT. This publication serves as the commander's primary guide for implementing the OTP.

ANNUAL PROFICIENCY

2-21. Annual proficiency is the primary way by which commanders maintain readiness in their SUAS Operator Training Program. The first annual proficiency evaluation must be completed within 12 months of a SUAS operator's mission certification. Subsequent annual proficiencies are required every 12 months thereafter. The annual proficiency period is a three-month window that units align with their fiscal year training calendar to maximize training synchronization. It consists of a written examination, combat identification test, and a live flight evaluation. Annual proficiency evaluations are conducted by an MT or UT as follows:

- The SUAS operator's written examination is an open-book exam, approved by the unit commander. The written exam will consist of 50 questions based off of paragraph 2-22. A score of 80% is considered passing
- A combat identification (CID) test (see Chapter 4)
- The SUAS operator must demonstrate proficiency in the most difficult method of control for flight aligned with the unit's METL during the live flight evaluation. Operators must conduct an annual proficiency evaluation for every category of system that they are certified on based off the similar series SUAS Table 2-1.
- Evaluate any additional unit level tasks at the commander's discretion.

Annual Proficiency Evaluation Period Example

A unit has aligned its annual proficiency evaluation period (three-month window) based off of a brigade combined arms live-fire exercise occurring in August. The unit's annual proficiency window is now June through August to get all SUAS annual proficiency evaluations complete.

If an SUAS operator completes their mission certification within the annual proficiency three-month window, that operator's annual proficiency evaluation is considered complete for that fiscal year.

2-22. After the assessment, the MT or UT will debrief the individual and annotate the event in the Soldier's individual training folder or approved electronic record and ensure entries are reflected in the Army's designated training management system.

Note. Operators who fail to meet the annual proficiency standards will be processed in accordance with paragraph 2-6.

2-23. The examinee must have a working knowledge and understanding of all applicable topics in the respective subject areas. At a minimum, the MT or UT will select two topics from each appropriate subject area. If the examinees themselves are MTs or UTs, they will also demonstrate the ability to instruct and evaluate any topic.

- Regulations and publications (this publication, AR 95-1, local SOPs, and regulations), which includes:
 - OTP requirements.
 - Unit training plan requirements.
 - SOP requirements.
 - Weather restrictions and procedures.
 - Local airspace usage.
 - Publications required for using the SUAS.
 - Forms and records.
- Operating limitations and restrictions, which includes:
 - Airspeed limits.
 - Environmental restrictions.
 - Range limitations.
 - Other limitations.
- SUAS emergency procedures and malfunction analysis, which includes:
 - Emergency terms and their definitions.
 - Battery malfunctions.
 - Link loss.
 - No hub communication.
 - Global Positioning System loss.
- Tactical and mission tasks, which includes:
 - Mission statement and employment methods.
 - Terrain analysis.
 - Launch and recovery site selection.
 - Battlefield environment.
 - Fratricide prevention.
 - Tactical reports.
 - Fire support.
 - Downed SUAS procedures.
 - Tactical airspace use, control, and management (JP 3-52).
 - Tactical airspace coordination.
 - Laser operations.
 - Laser performance detractors (if required).

WAIVERS AND DEVIATIONS

2-24. All waivers and deviations will be in accordance with AR 95-1.

LOCAL AREA ORIENTATION

2-25. Local area orientations are essential prior to conducting flight training. The local area orientation ensures familiarity with local operating procedures, policies and areas of operation. An operator must complete a local area orientation before being designated Mission Certified. The local area orientation should include:

- Reviewing applicable maps.
- Reviewing local environmental conditions, limitations, and restrictions.
- Obtaining range and restricted-area information.
- Obtaining information on frequencies and access phone numbers.
- Review of airspace procedures in the local area.

Note. Army commands, particularly those operating near sensitive borders, should establish additional requirements or restrictions for local area orientations.

WARNING

Units must adhere to see and avoid requirements established in federal aviation regulations, AR 95-1, AR 95-2, and local airspace and range policy to maintain separation from other air traffic. If operating beyond visual line of sight outside of restricted, one or more ground based observers or chase aircraft may be required to meet federal aviation regulations to maintain visual contact with the unmanned aircraft or the surrounding airspace.

SELF-START PROVISION

2-26. Brigade-level commanders and above are required to approve self-start programs prior to execution. Self-start allows the commander the ability to train a task or start a training program in which the unit is not currently trained. Commanders are authorized to “self-start” their training program if specific implementation guidance is not otherwise addressed. Commanders will select the most experienced MT to conduct the self-start initial training on trainers using conditions, standards, and the descriptions as outlined in the tasks listed in the training support packages. UTs will not train the task until they have been successfully evaluated by an MT. All other duty designations must receive training and evaluation by the appropriate standardization personnel prior to conducting the task.

2-27. In the absence of an MT, and with the approval of the first O-6 in the chain of command, commanders will select the most proficient UT to conduct the self-start initial training and evaluations using the conditions, standards, and descriptions outlined in the tasks listed in the training support packages (TSP). This authorization is temporary and remains in effect only for the time required to complete the specific TSP, not to exceed 6 months.

2-28. Commanders may use the self-start provision if all the unit’s MTs and UTs have lost proficiency to reestablish currency. This provision does not apply if the operator has not flown within the last calendar year.

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Chapter 3

Initial Licensing Training

This chapter establishes the Army's standardized Initial License Training pathway for all SUAS operators. The purpose is to define the mandatory requirements, evaluation criteria, and standards for the complete six-table progression an individual must successfully execute to earn a baseline "SUAS Operator License." This framework establishes the foundational knowledge and flight skills required to produce a safe and competent operator. Adherence to this chapter ensures a consistent and measurable training process across the force, regardless of military occupational specialty or unit type, and serves as the strict prerequisite for advancing to Mission Certification training (Chapter 4) and tactical employment.

INITIAL LICENSE TRAINING TABLES

3-1. The Initial License Training tables (Table 3-1) provide a progressive, standardized method for training and evaluating SUAS operators. Administered by an MT or UT, this framework consists of a sequential, six-table structure (Tables I-VI) that seamlessly blends classroom academics, simulated practice, and live-flight execution. Operators must successfully complete each table to standard before advancing to the next, culminating in a comprehensive live-flight evaluation.

Table 3-1. Initial license training

<i>Table I Academics</i>	<i>Table II Pre-Flight Simulations</i>	<i>Table III Drills</i>	<i>Table IV Advanced</i>	<i>Table V Practice</i>	<i>Table VI Licensing</i>
• Academics • Hands-on Platform Instruction • Written Exam • System Specific	• Simulated ▪ Instruction ▪ Basic Tasks ▪ Advanced Tasks • Unit Specific Tasks	• Live Flight Instruction • Equipment Procedures • Live Basic Tasks • Unit Specific Tasks	• Live Flight Instruction • Equipment Procedures • Live Advanced Tasks • Unit Specific Tasks	• Live Flight Instruction • Equipment Procedures • Practice Live Basic and Advanced Tasks • Unit Specific Tasks	• License Flight Instruction • Equipment Procedures • Unit Specific Tasks • License Flight Evaluation Standard

Note. While this training circular establishes the mandatory baseline for licensing, units are authorized and encouraged to add "Unit-Specific Tasks" to Tables I-IV. These tasks should be tailored to the unit's organic platforms, payloads, and anticipated mission sets to begin building mission-relevant proficiency early in the training pipeline. Any unit-specific tasks integrated into these initial tables must be incorporated into the Table V practice and formally evaluated during the Table VI Licensing flight.

TABLE I - ACADEMICS

3-2. Table I is the classroom portion of operator training and includes mandatory and unit-specified academic instruction, hands-on platform instruction, and a written examination. Classroom instruction is conducted sequentially and covers a standard minimum list of required topics:

- Academics
- Hands-on Platform Instruction
- Written Exam

Academics

3-3. Commanders, MTs, and UTs will ensure operators receive instruction on the mandatory minimum academic topics listed in Table 3-2. This instruction provides the common academic baseline for the entire force. In addition to these minimums, commanders, MTs, and UTs will incorporate any COE or unit-specific academic subjects required for their unique platforms, payloads, and mission sets. All instruction should be tailored to the SUAS groups and capabilities organic to the unit.

Table 3-2. Licensing academics

<i>TRAINING CATEGORY</i>
<i>ACADEMICS</i>
<i>Regulatory</i>
Military Policy, Doctrine, and Directives
Airspace
Record Keeping
<i>Safety</i>
Army Safety
<i>Weather</i>
Effects of meteorology on unmanned aircraft systems & mission
Identify Weather Hazards
<i>Aerodynamics</i>
Flight Aerodynamics
Aerodynamic Effects
<i>Mission</i>
Mission Planning
<i>Communications</i>
Signal Theory
Data Links
<i>Aircraft Operations</i>
Aircraft Systems
Emergency Procedures

Hands-on Platform Instruction Requirements

3-4. The hands-on platform instruction serves as the critical bridge between academic theory and practical application. It provides the operator with their first opportunity for hands-on familiarization with the SUAS components they will be required to operate and be evaluated on throughout the Initial Licensing Training pathway. The objective is to build an operator's confidence and procedural knowledge through physical identification, set up, and manipulation of the actual hardware ensuring a seamless transition to flight training. The following must be conducted as applicable to SUAS platforms:

- System components and characteristics:
 - Identify all major hardware components of the airframe, controller, and support equipment.
 - Describe the basic platform characteristics and operational limitations
- Assembly and connections:
 - Perform pre-flight assembly of the airframe and associated components.
 - Establish all inter-device connections to include controller pairing and linking to external software (for example Android Team Awareness Kit, mapping applications).
- Payloads and power:
 - Mount and connect sensor payloads.
 - Describe the characteristics and limitations of organic sensor payloads.
 - Demonstrate proper battery handling, charging, and inspection procedures.
- Operator-level maintenance:

- Review basic pre-flight and post-flight maintenance tasks (propeller inspection/replacement, lens cleaning, system wipe down).
- Perform basic pre-flight and post-flight maintenance tasks.

Written Examination

3-5. The written examination is a comprehensive assessment of an operator's knowledge of the topics covered in the classroom portion. Unit specific information should be reserved for annual proficiency examination. Operators must receive an 80% in each topic to pass. The written examination must be at least 50 questions composed of:

- 40 x Academic questions
- 10 x System specific question

Note. The units MTs and UTs will advise OTP Commanders on all additional final exam questions and alternate exam options for approval.

Note. If qualifying on more than one system, increase the final exam questions by 5 for each additional similar series system and by 10 for each dissimilar series system. If an Operator has already completed the final exam but is licensing on a new system, they need to take and pass the 10 questions for that specific system.

Note. System specific questions should focus on the tactical employment of system capabilities.

TABLE II - PRE-FLIGHT SIMULATIONS

3-6. Table II is entry-level simulated flight training designed to introduce operators to flight fundamentals and systems control. This table progressively builds operator confidence and establishes the basic proficiency required for safe operations. The operator will execute all applicable tasks from Table 3-3, plus any unit-specific simulated tasks within an approved simulation environment. An operator must pass a simulated flight evaluation at the conclusion of this table before advancing to Table III.

Table 3-3. Small unmanned aircraft system training flights

<i>Tasks Level</i>	<i>Flight Tasks</i>	<i>Rotorcraft</i>	<i>VTOL</i>	<i>Fixed Wing</i>	<i>Type of Flight</i>
Basic	Takeoff	X	X	X	S/L
Basic	Hover	X	X		S/L
Basic	Level Flight	X	X	X	S/L
Basic	Accelerate	X	X	X	S/L
Basic	Decelerate	X	X	X	S/L
Basic	Employ System Capability	X	X	X	S/L
Basic	Land	X	X	X	S/L
Advanced	Box Drills	X	X	X	S/L
Advanced	Figure 8	X	X	X	S/L
Advanced	Orbit	X	X	X	S/L
Legend: L: live flight S: simulated flight VTOL: vertical takeoff and landing					

Pre-Flight Simulation Instructions

3-7. Utilizing an approved simulator, the MT or UT will instruct the operator on basic flight skills in a controlled, virtual environment. As the "crawl phase" of practical application, Table II is focused on developing an operator's ability to perform basic maneuvers and safely operate the platform; it is not designed

to accomplish tactical missions. The operator will conduct all mandatory tasks listed in Table 3-3 to standard. This is the first opportunity for MTs and UTs to add simulated tasks tailored to their unit's specific platforms or mission sets. For detailed tasks, conditions, and standards, see Appendix A.

Note. System capabilities include organic payloads (for example electro-optical, infrared, forward-looking infrared)

Note. While training on simulators, operators should attempt to utilize controllers similarly used during live flights if able.

Note. If utilizing multiple similar series SUAS platforms during licensing training, operators only need to execute Table II with one similar series platform within the synthetic training environment (if available).

Note. To ensure sustained operator competence and confidence it is highly encouraged that units conduct multiple iterations of Table II.

Note. When embedded or standalone simulation is unavailable for a specific platform, the unit will utilize other simulated options (such as Virtual Battlespace 3, commercial-off-the-shelf flight simulator software, or Reconfigurable Academic and Adaptive Digital training environment). It is essential that the selected simulator closely replicates the flight characteristics of the specific platform (for example rotorcraft, fixed wing, vertical takeoff and landing).

Simulator Considerations

3-8. A variety of simulation options, from organic and embedded trainers to commercial-off-the-shelf software, can be used to accomplish Table II. While the organic trainer for a specific SUAS is always an acceptable training device, commanders and MTs should evaluate any supplemental or standalone simulators based on their ability to provide the most realistic training. An ideal simulator should possess the capability to:

- Closely replicate the flight characteristics of the specific SUAS category (for example rotorcraft, vertical takeoff and landing, fixed wing).
- Allow the MTs and UTs to manipulate key environmental conditions (for example visibility, wind).
- Support various methods of control applicable to the platform (for example manual, stabilized, semi-autonomous).
- Allow practice of the tasks and sequences required for the live-flight tables III-VI.
- Provide basic flight metrics for after-action review.

Note. The primary goal of simulation is to prepare the operator for the entire live-flight licensing pathway. If a system's organic embedded simulator has limited features, it is still the primary training tool; however, units are highly encouraged to supplement it with more robust simulation software if available to enhance operator proficiency.

TABLE III - DRILLS

3-9. Table III marks the operator's transition from simulation to live flight. The primary purpose is twofold. First, it allows the operator to gain a firsthand feel for the aircraft's flight characteristics. Second, it serves to safely orient the operator to the training area and begin integrating payload operation with basic flight. This table focuses on demonstrating proficiency in the basic tasks listed in Table 3-3 to the standards outlined in Appendix A and allows units to add specific tasks relevant to their platforms.

TABLE IV - ADVANCED

3-10. Table IV builds upon the foundational skills of Table III by requiring the operator to execute complex, controlled maneuvers. The purpose is to evaluate an operator's ability to maintain precise aircraft control while performing the advanced tasks from Table 3-3. These patterns require the operator to manage

coordinated turns and maintain a specific ground track, demonstrating a higher level of flight skill. Additionally, units may incorporate specific advanced tasks relevant to their organic platforms into this table.

TABLE V - PRACTICE

3-11. Table V is the culminating practice event designed to ensure an operator is fully prepared for the final licensing flight. This table requires the operator to execute all mandatory basic and advanced tasks from Table 3-3, as well as any unit-specific tasks introduced in previous tables, across one or more training flights. The purpose is to build an operator's expertise under pressure by conducting these maneuvers to tighter tolerances and at a higher tempo than required for the actual licensing evaluation. Successful completion of this demanding proficiency flight indicates a high probability of success for the Table VI evaluation.

TABLE VI - LICENSING

3-12. Table VI is the culminating, formal evaluation that validates an operator has met all standards required for an SUAS Operator License. This evaluation requires the operator to execute a complete flight profile by combining all mandatory tasks from Tables III and IV, plus any required unit-specific tasks. The operator must demonstrate proficiency in all evaluated tasks to the prescribed standards detailed in Appendix A. An operator who successfully completes Table VI is awarded their SUAS Operator License and is authorized to advance to Mission Certification training.

EMERGENCY PROCEDURES

3-13. Emergency procedures are designed to address abnormal or emergency situations that could reasonably be encountered during flight operations. The primary objective of these procedures is to mitigate risks to life, limb, or eyesight while ensuring the safety of personnel, equipment, and the surrounding environment. During all emergencies, priority must be given to protecting personnel and preventing harm.

LIMITATIONS

3-14. Operators must adhere to SUAS limitations as specified in the airworthiness release and the operator's manual at all times. Operating the SUAS outside of its prescribed limitations is strictly prohibited, except in emergency situations where there is an imminent risk to life, limb, or eyesight. Any operation beyond these limitations, outside of such emergencies, is deemed unsatisfactory.

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Chapter 4

Mission Certification Training

This chapter establishes the Army's standardized Mission Certification training pathway for training and evaluating licensed SUAS operators in tactical employment. The purpose is to build upon foundational flight skills, developing the tactical proficiency required for mission execution through a mandatory, progressive six-table structure. This framework encompasses mission-level academics, simulated rehearsal, and the live-flight execution of reconnaissance, surveillance, and tactical reporting tasks. Completing this pathway ensures operators are validated to perform tasks that directly support their unit's METL and earn the designation of Mission-Certified.

MISSION PREPARATION

4-1. Upon entering the Mission Certification pathway (Chapter 4), a licensed operator begins mission preparation. This stage encompasses the progressive training of Mission Certification Tables I through V. During mission preparation, the operator learns and practices the mandatory mission level academics and tactical flight tasks. This phase is designed to build proficiency and culminates in a demanding practice evaluation that ensures the operator is fully prepared for their final certification flight.

MISSION CERTIFIED

4-2. An operator achieves the Mission Certified designation upon successful completion of Mission Certification Table VI, the formal evaluation flight detailed in Chapter 4. This status validates that the operator has demonstrated proficiency in all required tasks and is fully prepared to support their unit's METL. Upon designation, the operator is authorized to participate in collective training and combat operations and incurs the responsibility to adhere to all currency and standardization program requirements.

MISSION CERTIFICATION TABLES

4-3. The Mission Certification tables (Table 4-1) are the Army's progressive, standardized method for training and certifying licensed SUAS operators for tactical employment. Administered by a UT/MT, this framework consists of a complete six-table structure.

Table 4-1. Mission certification table

<i>Table I Academics</i>	<i>Table II Pre-Flight Simulations</i>	<i>Table III Drills</i>	<i>Table IV Advanced</i>	<i>Table V Practice</i>	<i>Table VI Mission Certification</i>
• Mission Level Academics • Unit Specific Tasks • Written Exam • Combat Identification	• Simulated Instruction • Fundamental Payload Instruction • Simulated Flight and Mission Training with Payload • Unit Specific Tasks	• Live Flight and Mission Instruction • Equipment Procedures • Live Flight and Mission Training: Area Recon, Route Recon, and Aerial Observations • Unit Specific Tasks	• Live Flight and Mission Instruction • Equipment Procedures • Live Flight and Mission Training: Transmit a SPOT Report and Call for and Adjust Indirect Fire • Unit Specific Tasks	• Live Flight and Mission Instruction • Equipment Procedures • Practice Evaluation Training on all applicable tasks • Unit Specific Tasks	• License Flight and Mission Instruction • Equipment Procedures • Unit Specific Tasks • Mission Certification Flight Evaluation Standard

Note. "Fundamental Payload Instruction" within Table II is dedicated exclusively to the utilization of payloads that are not organic to the SUAS platform. Training must focus on how to operate and tactically employ these specific mission-enhancing systems as they differ from the baseline payloads used during initial licensing. Examples of these payloads include, but are not limited to gimballed intelligence, surveillance, and reconnaissance sensors; electronic warfare packages; communications relay packages; chemical, biological, radiological, and nuclear detection payloads; or inert lethal payloads.

Note. While this training circular establishes the mandatory baseline for mission certification, units are authorized and encouraged to add "Unit-Specific Tasks" to Tables I-IV. These tasks should be tailored to the unit's organic platforms, payloads, and anticipated mission sets to begin building mission-relevant proficiency early in the training pipeline. Any unit-specific tasks integrated into these initial tables must be incorporated into the Table V practice flight and formally evaluated during the Table VI Mission Certification flight.

TABLE I - ACADEMICS

4-4. Table I is the classroom portion of Mission Certification. Instruction covers mission-level academics, unit-specific METL-focused lessons, a written exam, and a CID evaluation. To progress past Table I, an operator must demonstrate knowledge of mission planning and pass the CID test.

Academics.

4-5. Commanders, MTs, and UTs will ensure operators receive instruction on the mandatory minimum academic topics listed in Table 4-2. This baseline instruction provides the common tactical foundation required for all SUAS operators across the force.

Table 4-2. Mission level training category

MISSION LEVEL TRAINING CATEGORY	
MISSION LEVEL ACADEMICS	
Perform Mission Route Selection and Analysis	A
Perform Aerial Navigation and Dead Reckoning	A
Understanding Tactical Mission Reporting	A
Understand and Apply Fire Support and Airspace Coordination Measures	A
Understand the Fundamentals of Reconnaissance and Security	A
Perform Map Analysis of the Mission Operations Area	A
MISSION PROGRESSION TASKS	
Perform Route Reconnaissance	S/L
Perform Area Reconnaissance	S/L
Perform Aerial Observation	S/L
Transmit a SPOT Report	S/L
Call for and Adjust Indirect Fire	S/L
Legend A: academics L: live flight S: simulated flight	

Unit-Specific Lessons

4-6. In addition to the baseline subjects, commanders will add academic topics tailored to their unit's specific METL, organic platforms, and anticipated operational environment. This localized instruction is critical for transitioning an operator from general tactical knowledge to specialized mission execution. MTs and UTs must develop lesson plans that integrate the SUAS into the unit's scheme of maneuver.

Written Exam

4-7. The written examination is a comprehensive assessment of an operator's knowledge of the topics covered in the classroom portion. Unit specific information should be reserved for annual proficiency examination. Operators must receive an 80% in each topic to pass. The written examination must be at least 25 questions composed of at least three questions per academic topic.

Combat Identification Evaluation

4-8. The CID test measures the ability of an operator to identify targets. Conditions should simulate environment based on current equipment capabilities (for example day, night, infrared/thermal). The CID test is essential in preparing new operators to quickly identify friendly and enemy units. The test consists of the following at minimum:

- 30 images (10 of which must be friendly and 15 of which should be shown under infrared/thermal, if equipped).
- Images should be presented from the appropriate tactical perspective.

4-9. Examination Procedures are as follows:

- Prepared with a test sheet, operators are shown images on a main screen.
- Operators have ten seconds to identify the images by the North Atlantic Treaty Organization designation or name and record them on their test sheet.
- After ten seconds the next image is shown. Repeat until all 30 images are presented.
- Operators must receive an 80% to pass
- Failure to properly identify friendly equipment is an automatic failure.:

Note. To ensure the validity of the assessment, MTs must develop multiple versions of the CID test in accordance with test construction best practices. An operator must not be administered the same version of the CID test during Mission Certification that they could complete during future training phases or evaluations (for example annual proficiency or Gunnery).

Note. Units employing SUAS in a non-tactical role (for example Corps of Engineers dam inspections, research and academic activities within Army research laboratories; Research, Development, and Engineering Command; military academies; or public affairs events) are not required to conduct the CID requirements. If mission use changes to a tactical role (for example recon or target acquisition), operators must pass the CID requirement prior to mission execution. The MT will document the change, conduct the training, and record the results.

TABLE II - PRE-FLIGHT SIMULATIONS

4-10. Utilizing an approved simulator, the MTs and UTs will instruct the operator on mission-oriented tasks in a controlled, virtual environment. Table II focuses on practicing the cognitive and procedural steps of tactical employment, such as reconnaissance patterns and tactical reporting, without the risk of live flight. The operator will conduct all mandatory Mission Tasks listed in Table 4-2, and any unit-specific simulated tasks to standard before advancing. For detailed tasks, conditions, and standards, see Appendix B.

TABLE III - DRILLS

4-11. Table III transitions the operator from mission simulation to mission-focused live flight. The purpose of this table is to practice individual mission tasks in a live environment. Operators will execute aerial observation, route reconnaissance, and area reconnaissance from Table 4-2 and any unit-specific tasks as standalone drills to build proficiency and confidence in performing them with a live aircraft. This ensures the operator can successfully execute a single tactical task before being required to link them into complex scenarios.

TABLE IV - ADVANCED

4-12. Table IV requires the operator to begin integrating multiple, distinct mission tasks into a single, cohesive flight. The purpose is to evaluate the operator's ability to transition between tasks and manage the

workflow required during a tactical mission profile. In this table, operators will incorporate tactical communications and fires integration by executing “Transmit a SPOT report” and “Call for and Adjust Indirect Fire” tasks in conjunction with the reconnaissance and unit-specific tasks learned in Table III (for example performing a route reconnaissance that culminates in calling for fire). This table builds the operator's capacity to manage multiple objectives and prepares them for the full mission profile in the practice and certification tables.

TABLE V - PRACTICE

4-13. Table V is the culminating practice event designed to ensure an operator is fully prepared for their final mission certification evaluation. This table requires the operator to plan and execute a complete, METL-based mission scenario by combining reconnaissance, reporting, and fires tasks from Tables III and IV along with any required unit-specific tasks. The purpose is to build expertise by managing the full mission workflow to tighter tolerances and at a higher tempo than required for the actual certification evaluation.

TABLE VI - MISSION CERTIFICATION

4-14. Table VI is the culminating, formal evaluation that validates a licensed operator is proficient in the tactical employment of the SUAS in support of their unit's METL. This evaluation requires the operator to plan and execute a complete, dynamic mission scenario from start to finish. The operator must demonstrate proficiency in all evaluated tasks to the prescribed standards detailed in Appendix B. An operator who successfully completes Table VI is awarded the designation Mission Certified and is authorized to participate in collective training and combat operations. A Mission Certified operator has a requirement to adhere to currency requirements and the standardization program.

Chapter 5

Gunnery Qualification Training

This chapter establishes the Army's standardized SUAS Gunnery Qualification training pathway. The purpose is to build upon the tactical proficiency of a Mission Certified operator, developing the specialized skills and evaluation standards required to safely and effectively employ weaponized or lethal-capable SUAS platforms during live-fire operations. It provides commanders with a standardized yet flexible methodology, establishing a sequential, mandatory six-table framework that defines the minimum requirements for qualification. This approach ensures Army-wide standardization while empowering leaders to tailor training scenarios to their specific organic platforms, lethal payloads, and unique mission requirements.

GUNNERY TRAINING PROGRAM

5-1. The commander is responsible and must ensure that the requirements under the gunnery training program are met. Gunnery Tables I-VI included in this chapter serve as templates that units can adapt to conduct gunnery training tailored to their specific operational needs.

5-2. The battalion commander and MT use the findings of the commander's assessment to determine the unit gunnery training strategy.

5-3. The commander's assessment is continuous and critical to a successful program. A Gunnery Program must be designed to sustain proficiency and increase lethality. The Master Trainer is the program manager. If any questions arise the program manager is highly encouraged to visit the AVCOE Directorate of Training and Doctrine Reconnaissance, Attack, Gunnery, Employment Branch SharePoint.

5-4. The battalion commander will develop a twelve-month gunnery plan titled "Unit Gunnery Training Strategy Memorandum". This plan must be published in memorandum format, signed by the commander, and made available for review by subordinate commands. This document will be kept for historical purposes for 24 months from its signature. The memorandum is an inspectable part of the Gunnery Program and must include the following:

- The unit's current strengths and weaknesses.
- Specific training objectives.
- Program milestones.
- Area of operations threat guidance.
- All planned GTs.

5-5. The commander determines training priorities by comparing unit readiness with the gunnery training standards. Areas include:

- Previous gunnery after action reviews.
- Gunnery no-notice evaluations.
- Individual experience levels.
- Table progression density, such as percentage of table qualified operators in proper modified table of organization and equipment positions.
- Single operator proficiency:
 - Gunnery Table Unit summaries.
 - Individual crew gunnery score sheets.
 - After action reviews.
- Collective training proficiency:
 - Prior collective assessments.
 - Combined training centers after action reviews.
- Training resource availability.
- Calculated "T Days," and scheduled resources. This step may require reprioritization of events on the long-range training calendar. Considerations are:

- Ranges.
- Ammunition.
- Taskings to subordinate units and requests for support from higher headquarters.
- Aircraft type and hours required to meet training objectives.

GUNNERY EVALUATOR

5-6. The following duty positions can perform gunnery evaluator duties:

- Master Trainers
- Unit Trainers

MINIMUM PROFICIENCY LEVELS

5-7. The minimum proficiency level (MPL) table (Table 5-1) is a list of mandatory common task requirements that all COEs must implement along with all addition unit specific gunnery tasks into their Gunnery Table scenarios. Engagements will be performed utilizing a specific training environment that supports the unit's METL.

Table 5-1. Minimum proficiency level tasks

<i>Task Number</i>	<i>Tasks Title</i>
G1	Provide a SPOT Report
G2	Call for and Adjust Indirect Fire
G3	Engage Stationary Target
G4	Engage Moving Target
G5	Conduct a Target Handover

Note: If employing any type of lethal weapon system an operator must be licensed and mission certified on that SUAS platform and weapon system.

Note: If the Gunnery Table VI evaluation was completed using a specific series, make, or model of a weapons employment system, the resulting Gunnery Qualification is valid only for that specific SUAS platform.

GUNNERY QUALIFICATION TABLES

5-8. The gunnery program outlined in this chapter is built upon a mandatory six-table framework (Table 5-2) that must be executed in sequential order to achieve Gunnery Qualification. This structure provides a standardized, progressive pathway required for all operators employing weaponized or lethal-capable SUAS across the Army.

Table 5-2. Gunnery qualification tables

<i>Table I Academics</i>	<i>Table II Pre-Flight Simulations</i>	<i>Table III Drills</i>	<i>Table IV Basic</i>	<i>Table V Practice</i>	<i>Table VI Gunnery Qualification</i>
• Academics • Unit Specific Tasks • Written Exam • Combat Identification	• Simulated Instruction • Simulated Flight Training • Introduction: Tactical Payload Skills (Lethal) • Tactical Movements and Communications • Unit Specific Tasks	• Equipment Procedures • Lethal Systems and Capabilities • Alignment, Harmonization, and Zeroing • Weaponneering Skills • Unit Specific Tasks	• Equipment Procedures • Live Tactical Engagements Instruction and Execution • Basic Weaponneering Skills • Unit Specific Tasks	• Equipment Procedures • Live Tactical Engagements Instruction and Execution • Advanced Weaponneering Skills • Unit Specific Tasks	• Equipment Procedures • Qualification Flight Evaluation Standard • Live Tactical Engagements Qualification • Unit Specific Tasks

5-9. Recognizing the rapid proliferation of SUAS platforms and the continuous development of new payloads, this training circular is intentionally not overly prescriptive. While each table outlines the minimum mandatory requirements and objectives (the "what"), it grants COEs and units flexibility to design and adapt the specific training scenarios (the "how"). This empowers leaders to tailor the execution to their organic systems, payloads, available munitions, and unique battlefield employment concepts. This approach ensures every operator meets a common Army standard of gunnery proficiency while enabling COEs to create experts optimized for their specific mission requirements.

5-10. Gunnery Tables I-VI apply to all SUAS Groups 1-2 that are weaponized and/or capable of lethal effects. Gunnery Tables I-VI will be conducted to meet the intent of the unit's METL.

5-11. Gunnery Tables I-VI are an annual requirement. Unlike Initial Licensing and Mission Certification, the similar series SUAS framework does not apply to gunnery. An operator qualified on multiple weaponized or lethal-capable SUAS platforms must successfully complete the annual Gunnery Tables I-VI requirements for each specific platform on which they maintain a gunnery qualification.

GUNNERY TABLE I - ACADEMICS

5-12. Gunnery Table I establishes the academic foundation required for all SUAS operators. It consists of four components: academic instruction, unit specific lessons, a written examination, and a CID test.

Academic Instruction:

5-13. Commanders, MTs, and UTs will ensure operators receive instruction on the mandatory minimum gunnery topics listed in Table 5-3. This instruction provides the common academic baseline for the Army. In addition to these minimums, commanders will incorporate any COE or unit-specific academic subjects required for their unique platforms, payloads, and mission sets. All instruction should be tailored to the SUAS groups and capabilities organic to the unit.

Table 5-3. Gunnery academics

<i>Gunnery Academics</i>
Fratricide Prevention
Employment Methods
Mission Equipment
Call for Indirect Fire
Tactical Reports
Munition Minimum and Maximum Ranges

Unit-Specific Lessons

5-14. In addition to the baseline subjects, commanders will add academic topics tailored to their unit's specific METL, organic weaponized platforms, and anticipated operational environment. This localized instruction is critical for transitioning an operator from general tactical knowledge to specialized lethal execution. MTs and UTs must develop lesson plans that integrate the weaponized SUAS into the unit's scheme of maneuver and fire support plan.

Written Exam

5-15. The written examination is a comprehensive assessment of an operator's knowledge of the topics covered in the classroom portion. Operators must receive an 80% in each topic to pass. The exam must cover the mandatory topics as well as any additional unit-required subjects. The written examination must be at least 25 questions composed of at least three questions per academic topic.

Combat Identification

5-16. The CID test measures the ability of an operator to identify targets. Conditions should simulate environment based on current equipment capabilities (for example day, night, infrared/thermal). The CID

test is essential in preparing new operators to quickly identify friendly and enemy units. The test consists of the following at minimum:

- 30 images (10 of which must be friendly and 15 of which should be shown under infrared/thermal, if equipped).
- Images should be presented from the appropriate tactical perspective.

5-17. Examination Procedures are as follows:

- Prepared with a test sheet, operators are shown images on a main screen.
- Operators have ten seconds to identify the images by the North Atlantic Treaty Organization designation or name and record them on their test sheet.
- After ten seconds the next image is shown. Repeat until all 30 images are presented.
- Operators must receive an 80% to pass
- Failure to properly identify friendly equipment is an automatic failure.:

Note. To ensure the validity of the assessment, MTs must develop multiple versions of the CID test in accordance with test construction best practices. An operator must not be administered the same version of the CID test during Gunnery Qualification that they completed during previous training phases or evaluations (for example annual proficiency or Mission Certification).

GUNNERY TABLE II - PRE-FLIGHT SIMULATIONS

5-18. Gunnery Table II is a performance-based evaluation in a simulated environment that serves as the gateway to live training. The purpose of this table is to certify that an operator can correctly execute all common tasks from the MPL table (Table 5-1) and any unit specific tasks to Army standards before consuming live resources. The MT will develop a scenario based on the unit's METL that requires the operator to demonstrate proficiency in these MPL tasks. Successful completion of GT II validates the operator has the fundamental skills required to safely progress to live-flight operations.

GUNNERY TABLE III - DRILLS

5-19. Gunnery Table III marks the operator's transition from simulation to live flight with munitions on both common tasks and unit specific tasks. The primary purpose of this table is twofold. First, it allows the operator to gain a feel for the SUAS flight characteristics with the added weight and aerodynamic effects of the payload. Second, it serves to safely orient the operator to the live-fire range, including all airspace boundaries, altitudes, flight routes and corridors, and target arrays. This table provides the necessary repetitions to build confidence and procedural fluency in the live environment before advancing to qualification tables.

GUNNERY TABLE IV - BASIC

5-20. Gunnery Table IV serves as the operator's entry into live-fire training. Its purpose is to train and evaluate an operator's ability to execute all mandatory MPL tasks and any unit-designated basic gunnery tasks to standard. This table requires the operator to safely and effectively engage static targets under optimal conditions, providing the commander with the initial validation that the operator is competent in employing their organic SUAS platform and payload.

GUNNERY TABLE V - PRACTICE

5-21. Gunnery Table V is an intermediate, live-fire event that builds upon the skills from GT IV. This table reinforces proficiency in MPL and unit tasks while systematically increasing tactical complexity. The focus is on preparing an operator for qualifications by introducing more demanding conditions such as multiple targets, varied target exposures, or scenarios requiring basic tactical maneuvering. GT V is designed to bridge the gap between basic proficiency and advanced employment by developing the operator's tactical expertise under pressure.

GUNNERY TABLE VI - GUNNERY QUALIFICATION

5-22. Gunnery Table VI is the culminating qualification event that validates an operator is proficient in all aspects of SUAS gunnery. Conducted on an approved course, this table evaluates an operator's mastery of all MPL and unit-specific tasks by presenting complex and time-constrained scenarios. An operator who

successfully completes Gunnery Table VI is considered Gunnery Qualified and is authorized to participate in collective and integrated live-fire events.

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Chapter 6

Task Development

AVCOE, as the proponent for SUAS training, establishes the universal, baseline standards that all operators must meet in accordance with this training circular. It is incumbent upon all other COEs to build upon this foundation. Each COE is responsible for analyzing their unique METL and developing the additional, specialized tasks required to ensure their operators are fully prepared for their specific operational environments. To develop an additional task, the COEs will create the task in the format described in this chapter; assign an individual, tactical, or collective series number to the task; and add it to the critical task list along with iteration and mode requirements. Tasks will be developed in accordance with Army guidance and this training circular.

TASK MODEL DEVELOPMENT

6-1. When a unit develops an additional task, the commander must perform a risk analysis for performance of the task and determine the training required for personnel to attain proficiency in the task. The commander will ensure Soldiers receive the necessary academic and deployment of SUAS training for this new task during mission-level progression and will determine if there is a requirement for an annual qualification of the task. The additional task(s) must include the following:

- Task number and title.
- Conditions for task performance.
- Standards for task performance.
- Description of task performance.
- Considerations for task performance (such as environmental and safety).
- Requirements for training or qualification.
- References.

TASK FORMAT

6-2. The following format will be used to develop all additional individual, tactical, and collective series tasks (see Figure 6-1, page 28). Each task element is explained further in the following paragraphs.

Note. The following task is intended to explain formatting style only; it does not constitute doctrinal procedure.

TASK XXXX (Individual, Tactical, Collective Series) Task Title (Perform Flight Mission Planning) CONDITIONS: The condition statement specifies the circumstances under which the task must be performed and identifies an initiating cue or triggering circumstance that lets the operator know how to accomplish the task. It also lists materials, equipment, and personnel required for task accomplishment. STANDARDS: The individual task standard describes the minimum acceptable level of performance in the field to successfully accomplish the task under the prescribed conditions. It notes how well someone should perform the task to be considered competent. For SUAS deployment tasks, standards are based on ideal conditions. DESCRIPTION: The description explains the preferred method for conducting a task to meet the standards. This manual cannot address all situations or alternate procedures/methods to accomplish a task. Tasks may be accomplished using other methods, as long as the task is done safely and the standards are met. Task descriptions are the “how to” portion of the task: • Descriptions will normally be divided into two sections: operator actions and procedures. • Ensure that the correct designation for the operator is used in the description to avoid confusion. • Procedures identify the preferred method of accomplishing the task. • Make sure the standards for the task are clearly defined in the “Standards” section; however, it may be necessary to refer the reader to the description section for specific requirements. • Deviations from task procedures—but not operator actions—are authorized only if task standards and safety are not compromised. ENVIRONMENTAL CONSIDERATIONS: This section of a task must address the unique requirements of performing the task under different flight modes or under adverse environmental conditions. The following are examples of environmental considerations: • Altitude, apparent ground speed, and rate of closure are difficult to estimate at night. • Surrounding terrain or vegetation may decrease contrast and degrade depth perception during the approach. Before descending below obstacles, determine the need for artificial lighting. • Use proper scanning techniques to avoid spatial disorientation. • Acquire, identify, track, and/or designate targets using the sensor’s optimum capabilities (EO, IR) for a given situation based on METT-TC (I) mission, enemy, terrain and weather, troops and support available, time available, civil considerations, and informational considerations. TRAINING AND EVALUATION REQUIREMENTS: Training and evaluation requirements define whether the task will be trained and/or evaluated in the aircraft, simulator, or academic environment. Listing aircraft and/or simulator under the evaluations requirements does not preclude the evaluator from evaluating elements of the task academically to determine depth of understanding or planning processes. Some task procedures allow multiple methods to achieve standards. 1. Training may be conducted academically, with hands-on performance, in a simulator or live aircraft. 2. Evaluation will be conducted academically, with hands-on performance, in a simulator or live aircraft. REFERENCES: The reference section of the task lists the sources of information relating to that specific task. All official Army documents referenced for creation of the task need to be listed and any unofficial documents can be listed but are not required.
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Figure 6-1 Task template

TASK NUMBER

6-3. Task numbers are written in uppercase, bold format and run sequentially.

TASK TITLE

6-4. The task title describes the performance required of the Soldier on the job. It is frequently referred to as the task. The task title has one action verb, one object, and may also have a qualifier that describes the required action. Task titles are written in title case and bolded format. Acronyms will not be used in the title. Using standard, well-defined verbs-

- Provides and promotes clarity.
- Allows analysts, task selection boards, trainers, and Soldiers to understand what the task title means.
- Prevents duplication. Using standard verbs makes it simple to group tasks by verbs to avoid duplication.
- Promotes application of sound training principles.

CONDITIONS

6-5. A condition statement has two parts:

- A cue is a word, situation, or other signal for action for an individual to begin performing an individual task.
 - An internal cue is a signal to go from one element of a task to another.
 - A terminating cue indicates task completion.
- Descriptive data is information that identifies when, why, and where the task is performed and what resources (materials, personnel, and equipment) are required to perform the individual task.

6-6. The individual task condition statement is in standard paragraph format, containing one or more sentences. The following guidelines and tips should be used for writing an individual task condition statement:

- Identify the cue.
 - The cue may be evident or "understood" when writing a condition statement and may not require detail.
 - Specifically identify the cue if it is not evident. Identifying the cue may require studying items, such as organizational diagrams; mission analysis; threat information; actions performed by outside units, Soldiers, leaders, or events; or procedural manuals.
- Identify or describe the physical setting or the site of individual task performance. The amount of detail provided varies based on the effect that the setting has on the task performance.

Notes

6-7. Notes are included to provide clarification - to offer additional information, emphasize a point, or explain a procedure in more detail without being critical to safety (like a Warning) or preventing damage (like a Caution). They enhance understanding and ensure proper application of the technical data.

- The setting should not be too generic or too specific. The task is written to field performance, not training performance.
- When the task is performed at multiple performance sites, describe all sites as practicable.

Special Considerations

6-8. A special condition is an aiding or limiting factor that occasionally occurs and affects a Soldier's ability to perform the task to the established standard. These special conditions include, but are not limited to, wearing of mission oriented protective posture 4, night vision devices, or self-contained breathing apparatus when performing the task. These unique circumstances are identified as separate special condition statements when conducting the individual task analysis and are also entered under the conditions tab in the Combined Arms Center-approved automated development system.

6-9. Once changes to the task performance standard caused by performing the task under a special condition are identified, developers must include them. When writing a special condition statement, be aware that:

- More than one special condition simultaneously may affect task performance.
- A special condition may affect such standards as speed or accuracy.

STANDARDS

6-10. The individual task standard describes the minimum acceptable level of performance in the field to successfully accomplish the task under the prescribed conditions. It notes how well someone should perform the task to be considered competent. For SUAS deployment tasks, standards are based on ideal conditions.

6-11. Standards must be:

- Written in present tense.
- Written in standard paragraph format. The paragraph may contain one or more sentences and may include subparagraphs and/or bullets.
- Written in job holder language.
- Observable, measurable, achievable, objective, valid, reliable, usable, comprehensive, discriminating, and quantifiable.

6-12. A standard statement has two parts – performance and criterion:

- Performance is a verb phrase that identifies what action the standard will evaluate – that is the process the Soldier performs, the product produced, or a combination of both.
 - A process standard describes the critical task elements necessary for adequate task performance.
 - A product standard describes the end result of individual task performance. Product standards should be used when the process it takes to perform the task is not important, as long as the product (end result) is correct.
 - A combination standard is used when task performance produces both a product and process.
- Criterion may include, but is not limited to, accuracy, quantity, speed, and quality. Table 6-1 addresses recommended criteria for each type of task performance.

Table 6-1. Task performance criteria

<i>Individual Task Standard</i>	<i>Criteria to Use</i>
Product Standard	Accuracy, tolerances, completeness, format, clarity, number of errors, and quantity.
Process Standard	Sequence, completeness, accuracy, and speed of performance.
Combination Standard	Accuracy, tolerances, completeness, format, clarity, number of errors, quantity, sequence, and speed of performance.

6-13. Parts of example standard statements may include the following:

- Fire all 18 rounds (performance) and hit the target at least nine times (criterion).
- Camouflage the position (performance), so it cannot be detected from 35 meters forward (criterion).
- Calibrate the altimeter within 1-meter (performance) in accordance with TM 1-1550-695-13&P (criterion).

Note. It is preferred not to use a standard that refers to another document; but, in the case of aircraft maintenance, it is mandatory to use the technical manual. Duplicating the technical manual causes extra work and serves no value since the technical manual is used when the task is performed.

DESCRIPTIONS

6-14. Task descriptions are the "how to" portion of the task:

- Descriptions are normally divided into two sections: operator actions and procedures.
- Designations for the operator are used correctly in the description to avoid confusion.
- Procedures identify the preferred method of accomplishing the task.
- Standards for the task are clearly defined in the “Standards” section; the description section is referenced as needed for specific requirements.
- Deviations from task procedures—but not operator actions—are authorized only when task standards and safety are not compromised.

ENVIRONMENTAL CONSIDERATIONS

6-15. The environmental considerations section of a task must address the unique requirements of performing the task under different flight modes or under adverse environmental conditions. The following are examples of environmental considerations:

- Altitude, apparent ground speed, and rate of closure are difficult to estimate at night.
- Surrounding terrain or vegetation may decrease contrast and degrade depth perception during the approach. Before descending below obstacles, determine the need for artificial lighting.
- Use proper scanning techniques to avoid spatial disorientation.
- Acquire, identify, track, and/or designate targets using the sensor's optimum capabilities (electro-optical, infrared) for a given situation based on mission variables.

TRAINING AND EVALUATION REQUIREMENTS

6-16. Training and evaluation requirements define whether the task will be trained or evaluated in the aircraft, simulator, or academic environment. Listing aircraft or simulator under the evaluations requirements does

not preclude the evaluator from evaluating elements of the task academically to determine depth of understanding or planning processes. Some task procedures allow multiple methods to achieve standards.

REFERENCES

6-17. The reference section of the task lists the sources of information relating to that specific task. All official Army documents referenced for creation of the task must be listed. Any unofficial documents may be listed, but they are not required.

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Chapter 7

Individual Operator Training Record

The commander must ensure that an individual training folder is prepared and maintained for each SUAS operator in an operational duty position assigned or attached to the unit. The OTP records system provides commanders with a comprehensive performance record on each operator in their unit. The required forms can be completed by hand using dark blue or black ink or digitally. The forms are available from the Army Publishing Directorate website. SUAS operator training records are important quality control and standardization tools. They should be completed carefully, completely, and legibly.

REQUIREMENTS

7-1. Each operator will have an individual training folder either in two physical copies – one with the soldier and one with the commander – or in an approved electronic record. All major entries will be annotated in the individual training folder or approved electronic record and entered into the Army's designated training management system. The individual training folders will be maintained as follows:

- Each folder will serve as the operator's complete training history, containing physical copies of all relevant documents. These documents include, but are not limited to, qualifications, certifications, DA Form 7122 (*Crew Member Training Record*), DA Form 4507 (*Crew Member Grade Slip*), DA Form 4507-1 (*Maneuver/Procedure Grade Slip*) and any additional certifications (for example Master Trainer certification).
- DA Form 7122 will be used to document all individual operator's certifications and qualifications.
- DA Form 4507 and DA Form 4507-1 should be used to document training and evaluation flights.
- DA Form 4507 and DA Form 4507-1 will be filed on the right side of the individual training folder until training has been completed, and the event has been documented on DA Form 7122. Once the event has been entered on DA Form 7122, then DA Form 4507 and DA Form 4507-1 will be removed from the individual training folder.
- Operators performing SUAS duties will obtain their individual training folder and present it to the commander or the commander's representative within 14 days of arrival to the unit.

Note. Approved extensions or waivers will be retained in the miscellaneous section of the training folder until the end of the year.

7-2. Records should be recorded and maintained in accordance with TC 3-04.11.

Note. For all forms, when the instructions refer to "aircrew members" or other similar terminology, the term "operator" should be used in its place.

DEPARTMENT OF THE ARMY FORM 7122

7-3. DA Form 7122 is used to permanently record all individual operator evaluations and summaries. It is also used to record any change in operator status or other significant events.

7-4. Instructions for completing DA Form 7122 are as found on the form. For those instances where the instructions vary are found below:

- Aircraft. Enter the mission design series or Vendor name and model if a mission design series has not been assigned. If the event was performed solely in a flight simulator, enter the simulator designation (i.e. Virtual Battlespace 3, commercial-off-the-shelf flight simulator software, or Reconfigurable Academic and Adaptive Digital training environment)
- Event. Enter a short summary of the event on one line. Record events listed in TC 3-04.11 and those below:

- Start and completion of time-limited training programs, such as, initial qualification, mission preparation, mission certification, or MT certification.
- Start and completion of MT courses.
- Completion of significant training or retraining programs.
- MT designation.

7-5. Only the following events, in addition to those in TC 3-04.11, recorded on DA Form 7122 require the commander's signature:

- Mission certification
- Mission certification designation after failure of a hands-on skill assessment or a training deficiency.
- Return to previous duties after nonmedical suspension or mission certification re-designation.

DEPARTMENT OF THE ARMY FORM 4507

7-6. Instructions for completing DA Form 4507 are as found on the form. For those instances where the instructions vary are found below:

7-7. For "Aircraft Type," the mission design series or vendor name and model if a mission design series has not been assigned will be entered.

DEPARTMENT OF THE ARMY FORM 4507-1

7-8. Instructions for completing DA Form 4507-1 are as found on the form. For those instances where the instructions vary are found below:

7-9. For "Maneuver/Procedure," enter the task number followed by the task title as found in the METL and required by the unit's OTP. Units may list all tasks required by the commander's task list or develop separate forms for each training program.

Appendix A

Training Flight Tasks

A-1. The following outlines the requirements for conducting a simulated or live SUAS

A-2. flight. It provides the task, condition, and standard for operator performance and references Tables A-1 through A-9 that governs execution:

- Task: Conduct a simulated or live flight in accordance with the chart below.
- Condition: As a SUAS operator in a classroom or operational environment with a ground control station, aircraft, applicable manuals, this training circular, and local SOPs.
- Standard: Conduct a simulated or live flight in accordance with the chart below without safety violations and 100 percent go on all performance measures.

Table A-1. Tasks 1001-1003

Task Number	Task	Aircraft Type	Performance Measures	
1001	Takeoff	Rotorcraft	1. Increase throttle to provide sufficient power for Aircraft to lift off the ground.	
1002			VTOL	2. Ascend vertically in a controlled manner.
				3. Establish a stable hover at a safe altitude (approx. 10-20 feet AGL).
				4. Stabilize Aircraft in forward flight.
				5. Avoid any obstacles.
				6. Maintain situational awareness of Aircraft operation to ensure safety at all times.
		1003		Fixed Wing
2. Ascend vertically in a controlled manner.				
3. Ascend to a safe transition altitude while maintaining a stable hover				
4. Initiate transition to forward flight, ensuring a positive climb rate and airspeed gain				
5. Avoid any obstacles.				
6. Maintain situational awareness of Aircraft operation to ensure safety at all times.				
Legend: AGL – above ground level approx. – approximately VTOL – vertical takeoff and landing				

Table A-2. Tasks 1011-1023

1011	Hover	Rotorcraft	1. Maintain position over a designated ground reference point with minimal drift accounting for wind.		
1012			VTOL	2. Make smooth, minimal control inputs to maintain stability	
				3. Avoid any obstacles.	
				4. Maintain situational awareness of Aircraft operation to ensure safety at all times.	
		1013		Fixed Wing	1. Not Applicable
1021			Rotorcraft		1. Establish a hover.
					2. Adjust pitch and throttle setting to enter forward flight
					3. Maintain altitude throughout entire maneuver.
	4. Avoid any obstacles.				
	5. Maintain situational awareness of Aircraft operation to ensure safety at all times.				
1022	VTOL	1. Establish a hover.			
		2. Initiate the transition sequence to forward flight per the aircraft's technical manual (i.e., tilting rotors, engaging pusher motors).			
		3. As airspeed increases, manage pitch and throttle to maintain a constant altitude throughout the transition.			
		4. Avoid any obstacles.			
		5. Maintain situational awareness of Aircraft operation to ensure safety at all times.			
1023	Fixed Wing	1. Maintain straight and level flight.			
		2. Adjust pitch and throttle setting to increase speed.			
		3. Maintain altitude throughout entire maneuver.			
		4. Avoid any obstacles.			
		5. Maintain situational awareness of Aircraft operation to ensure safety at all times.			
Legend: i.e. – for example VTOL – vertical takeoff and landing					

Table A-3. Tasks 1031-1033

1031	Level Flight	Rotorcraft	1. Establish and maintain an assigned altitude while maintaining forward flight.
			2. Establish and maintain an assigned heading.
			3. Make smooth and timely corrections to maintain the intended direction of flight or flight path.
			4. Avoid any obstacles.
			5. Maintain situational awareness of Aircraft operation to ensure safety at all times.
1032		VTOL	1. Transition from hover to forward flight and climb to assigned altitude.
			2. Establish and maintain an assigned altitude and airspeed in forward flight.
			3. Make smooth and timely corrections to maintain the flight path and correct for deviations.
			4. Avoid any obstacles.
			5. Maintain situational awareness of Aircraft operation to ensure safety at all times.
1033		Fixed Wing	1. Establish and maintain an assigned altitude and airspeed in forward flight.
			2. Make smooth and timely corrections to maintain the flight path and correct for deviations.
			3. Avoid any obstacles.
			4. Maintain situational awareness of Aircraft operation to ensure safety at all times.

Legend:
VTOL – vertical takeoff and landing

Table A-4. Tasks 1041-1043

1041	Employ System Capabilities	Rotorcraft	1. Establish a stable hover or loiter that provides an optimal vantage point.
			2. Utilize the designated system/payload
			3. Manipulate sensor/payload controls (e.g., pan, tilt, zoom) to acquire the required information
			4. Verbally report or digitally transmit acquired information and resume flight.
			5. Avoid any obstacles.
			6. Maintain situational awareness of FPV Aircraft operation to ensure safety at all times.
1042		VTOL	1. Establish a stable hover or loiter that provides an optimal vantage point.
			2. Utilize the designated system/payload
			3. Manipulate sensor/payload controls (e.g., pan, tilt, zoom) to acquire the required information
			4. Verbally report or digitally transmit acquired information and resume flight.
			5. Avoid any obstacles.
			6. Maintain situational awareness of Aircraft operation to ensure safety at all times.
1043		Fixed Wing	1. Establish a stable hover or loiter that provides an optimal vantage point.
			2. Utilize the designated system/payload
			3. Manipulate sensor/payload controls (e.g., pan, tilt, zoom) to acquire the required information
			4. Verbally report or digitally transmit acquired information and resume flight.
			5. Avoid any obstacles.
			6. Maintain situational awareness of Aircraft operation to ensure safety at all times.
Legend: e.g. – for example FPV – first-person view VTOL – vertical takeoff and landing			

Table A-5. Tasks 1051-1053

1051	Decelerate	Rotorcraft	1. Maintain straight and level flight	
1052			VTOL	2. Adjust pitch and throttle setting to reduce speed.
				3. Establish a hover.
				4. Maintain altitude throughout entire maneuver.
				5. Avoid any obstacles.
				6. Maintain situational awareness of Aircraft operation to ensure safety at all times.
		1053		Fixed Wing
2. Initiate the transition sequence from forward flight to hover per the aircraft's technical manual.				
3. As airspeed decreases, manage pitch and power to maintain altitude and avoid an aerodynamic stall.				
4. Establish a stable hover.				
5. Avoid any obstacles.				
6. Maintain situational awareness of Aircraft operation to ensure safety at all times.				
1053		Fixed Wing	1. Maintain straight and level flight.	
			2. Adjust pitch and throttle setting to reduce speed.	
			3. Establish stable flight at a slower airspeed.	
			4. Maintain altitude throughout entire maneuver.	
			5. Avoid any obstacles.	
			6. Maintain situational awareness of Aircraft operation to ensure safety at all times.	

Legend:
VTOL – vertical takeoff and landing

Table A-6. Tasks 1061-1063

1061	Land	Rotorcraft	1. Pick a desired landing point.
			2. Reduce speed while orienting on desired landing point.
			3. Enter a hover.
			4. Using smooth control inputs, descend towards intended landing point.
			5. Land on intended landing point.
			6. Avoid any obstacles.
			7. Maintain situational awareness of Aircraft operation to ensure safety at all times.
1062		VTOL	1. Pick a desired landing point.
			2. Reduce speed while orienting on desired landing point.
			3. Approach the landing point and execute a smooth transition from forward flight to hover mode
			4. Perform a controlled vertical descent and land gently.
			5. Land on intended landing point.
			6. Avoid any obstacles.
			7. Maintain situational awareness of Aircraft operation to ensure safety at all times.
1063		Fixed Wing	1. Pick a desired landing point.
			2. Establish a stabilized final approach path into the wind
			3. Manage descent rate and airspeed for a safe approach.
			4. Flare the aircraft just above the ground to touch down within the designated landing zone.
			5. Avoid any obstacles.
			6. Maintain situational awareness of Aircraft operation to ensure safety at all times.
Legend: VTOL – vertical takeoff and landing			

Table A-7. Tasks 1071-1073

1071	Box Drills	Rotorcraft	1. Maintain straight and level flight.
			2. Begin a turn by inputting roll left or right in the direction of choice
			3. Apply yaw in direction of turn as needed to maintain nose of Aircraft or sensor in direction of turn
			4. Enter straight and level flight after completion of turn.
			5. Avoid any obstacles.
			6. Maintain situational awareness of Aircraft operation to ensure safety at all times.
			7. Complete two iterations of each direction of travel
1072		VTOL	1. Maintain straight and level flight at a constant altitude and airspeed.
			2. Begin a 90-degree turn by inputting roll in the desired direction to establish a bank.
			3. Apply pitch input (back pressure) as needed to maintain altitude throughout the turn.
			4. Roll out on the new heading to enter the next straight and level flight leg.
			5. Avoid any obstacles.
			6. Maintain situational awareness of Aircraft operation to ensure safety at all times.
			7. Complete the four legs and four turns of the box pattern.
1073		Fixed Wing	1. Maintain straight and level flight at a constant altitude and airspeed.
			2. Begin a 90-degree turn by inputting roll in the desired direction to establish a bank.
			3. Apply pitch input (back pressure) as needed to maintain altitude throughout the turn.
			4. Roll out on the new heading to enter the next straight and level flight leg
			5. Avoid any obstacles.
			6. Maintain situational awareness of Aircraft operation to ensure safety at all times.
			7. Complete the four legs and four turns of the box pattern.
Legend: VTOL – vertical takeoff and landing			

Table A-8. Tasks 1081-1083

1081	Figure 8	Rotorcraft	1. Maintain straight and level flight maneuvering towards the center point of the Figure-8.
2. Cross from left to right or vice versa at the center point.			
3. Apply yaw in direction of turn to orbit around corner point.			
4. Perform the same maneuver in the opposite direction.			
5. Avoid any obstacles.			
6. Maintain situational awareness of Aircraft operation to ensure safety at all times.			
7. Complete two full Figure-8 drills.			
1082		VTOL	1. Maintain straight and level flight, maneuvering to enter the first loop of the Figure-8.
			2. Initiate a coordinated, constant-radius turn by rolling into a bank around the first point.
			3. Apply continuous pitch and roll inputs as needed to maintain a constant altitude and circular ground track.
			4. As the aircraft approaches the center point, smoothly roll out and immediately roll into an opposite-direction bank to begin the second loop.
			5. Avoid any obstacles.
			6. Maintain situational awareness of Aircraft operation to ensure safety at all times.
			7. Complete two full Figure-8 drills.
1083		Fixed Wing	1. Maintain straight and level flight, maneuvering to enter the first loop of the Figure-8.
			2. Initiate a coordinated, constant-radius turn by rolling into a bank around the first point.
			3. Apply continuous pitch and roll inputs as needed to maintain a constant altitude and circular ground track.
			4. As the aircraft approaches the center point, smoothly roll out and immediately roll into an opposite-direction bank to begin the second loop.
			5. Avoid any obstacles.
			6. Maintain situational awareness of Aircraft operation to ensure safety at all times.
			7. Complete two full Figure-8 drills.
Legend: VTOL – vertical takeoff and landing			

Table A-9. Tasks 1091-1093

1091	Orbit	Rotorcraft	1. Establish a target to orbit.
			2. Orient the Aircraft on the target.
			3. Apply yaw in direction of orbit as needed to maintain nose of Aircraft or sensor on the target.
			4. Apply roll in opposite direction of orbit as needed to maintain direction of flight around target.
			5. Maintain target in FOV.
			6. Maintain altitude throughout entire maneuver.
			7. Enter straight and level flight after completion of orbit.
			8. Avoid any obstacles.
			9. Maintain situational awareness of Aircraft operation to ensure safety at all times.
			10. Complete two full orbits.
1092		VTOL	1. Identify a target or point of interest to orbit around.
			2. Initiate a coordinated turn toward the target by applying roll input to establish a consistent bank angle.
			3. Apply continuous pitch input (back pressure) as needed to maintain a constant altitude.
			4. Adjust bank angle and pitch as required to maintain a constant radius from the target.
			5. Maintain target in FOV.
			6. Maintain altitude throughout entire maneuver.
			7. Enter straight and level flight by rolling wings-level after completion of the orbit.
			8. Avoid any obstacles.
			9. Maintain situational awareness of Aircraft operation to ensure safety at all times.
			10. Complete two full orbits.
1093		Fixed Wing	1. Identify a target or point of interest to orbit around.
			2. Initiate a coordinated turn toward the target by applying roll input to establish a consistent bank angle.
			3. Apply continuous pitch input (back pressure) as needed to maintain a constant altitude.
			4. Adjust bank angle and pitch as required to maintain a constant radius from the target.
			5. Maintain target in FOV.
			6. Maintain altitude throughout entire maneuver.
			7. Enter straight and level flight by rolling wings-level after completion of the orbit.
			8. Avoid any obstacles.
			9. Maintain situational awareness of Aircraft operation to ensure safety at all times.
			10. Complete two full orbits.
Legend: FOV – field of view VTOL – vertical takeoff and landing			

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Appendix B

Mission Certification Tasks

B-1. The following outlines the requirements for conducting a simulated or live SUAS flight under the conditions specified for Tables B-1 and B-2. It provides the task, condition, and standard for operator performance and references the corresponding training chart that governs execution.

- Task: Conduct a simulated or live flight in accordance with the chart below.
- Condition: As a SUAS operator in a classroom or operational environment with a ground control station, aircraft, applicable manuals, this training circular, and local SOPs.
- Standard: Conduct a simulated or live flight in accordance with the chart below without safety violations and 100 percent go on all performance measures.

Table B-1. Tasks 2001-2023

Task Number	Task	Performance Measures
2001	Perform Area Reconnaissance	1. Receive a mission brief
		2. Launch mission aircraft and maneuver to briefed positions at the time directed in the mission order/brief
		3. Perform area reconnaissance as directed in the mission brief
		4. Report critical information obtained from the reconnaissance mission and submit applicable reports.
		5. Conduct reconnaissance handover per Commander's reconnaissance guidance (engagement/bypass criteria, and available endurance of reconnaissance assets)
		6. Complete the assigned reconnaissance mission.
2012	Perform Route Reconnaissance	1. Receive a mission brief
		2. Launch mission aircraft and maneuver to briefed positions at the time directed in the mission order/brief
		3. Perform route reconnaissance
		4. Report critical information obtained from the reconnaissance mission and submit applicable reports.
		5. Complete aerial route reconnaissance and security missions as directed or until relieved.
2023	Perform Aerial Observation	1. Receive a mission brief
		2. Navigate to the appropriate briefed position.
		3. Conducted the appropriate reconnaissance and security fundamentals necessary to accomplish the mission, as briefed.
		4. Detected a target utilizing the appropriate payload sensor.
		5. Identify the target.
		6. Locate the target.
		7. Report the target, as briefed.
		8. Performed actions to maintain an unobstructed payload to target line of sight.
		9. Maintain a proper standoff range from the objective, target(s), terrain and other aircrafts throughout the mission.
		10. Perform coordination actions to provide early and accurate warning to the protected force.

Table B-2. Tasks 2034-2055

2034	Transmit a SPOT Report	1. Identify necessary information required for the report.
		2. Prepare the SPOTREP IAW FM 6-99.
		3. Transmit the SPOTREP.
2045	Perform Call for Fire as an Unmanned Aircraft System (UAS) Operator	1. Locate possible target(s).
		2. Obtain positive identification of the target(s), as required.
		3. Perform a call for fire.
		4. Read back the message to observer precisely.
		5. Perform an adjust fire.
		6. Perform battlefield damage assessment
		7. Perform coordination actions to position the aircraft out of the line of fire and maintain an unobstructed payload to target LOS.
2055	Perform Target Handover	1. Perform target handover while using voice communications as the transferring element.
		2. Perform target handover while using voice communications as the receiving element.
		3. Perform target handover by providing friendly asset location, target location, and target description as the transferring element.
		4. Receive target handover acknowledgement from receiving element.
Legend: FM – field manual IAW – in accordance with LOS – line of sight SPOTREP – SPOT report		

Glossary

The glossary lists acronyms and terms with Army or joint definitions. Where Army and joint definitions differ, (Army) precedes the definition. The proponent publication for terms is listed in parentheses after the definition.

SECTION I – ACRONYMS AND ABBREVIATIONS

AAR	after action review
ADP	Army doctrine publication
AR	Army regulation
AVCOE	Aviation Center of Excellence
CID	combat identification
COE	center of excellence
DA	Department of the Army
DOD	Department of the Army
FM	field manual
HSA	hands-on skill assessment
IR	infrared
JP	joint publication
METL	mission-essential task list
MPL	Minimum proficiency level
MT	Master trainer
OTP	Operator Training Program
SOP	standard operating procedure
SUAS	small unmanned aircraft system
TC	training circular
TP	United States Army Training and Doctrine Command Pamphlet
UT	Unit trainer

SECTION II – TERMS

No terms associated with this publication.

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References

All URLs were accessed on 06 May 2026.

REQUIRED PUBLICATIONS

These documents must be available to intended users of this publication.

DOD Dictionary of Military and Associated Terms. 28 March 2026.

FM 1-02.1. *Operational Terms*. 28 February 2024.

RELATED PUBLICATIONS

These documents are cited in this publication.

JOINT PUBLICATIONS

Most joint publications are available online: <http://www.jcs.mil/Doctrine>.

JP 3-30. *Joint Air Operations*. 28 April 2025.

JP 3-52. *Joint Airspace Control*. 22 October 2022.

ARMY PUBLICATIONS

Most Army doctrinal publications are available online: <https://armypubs.army.mil/>.

AR 95-1. *Flight Regulations*. 22 March 2018.

AR 95-2. *Air Traffic Control, Airfield/Heliport, and Airspace Operations*. 31 March 2016.

AR 350-1. *Army Training and Leader Development*. 01 June 2025.

ADP 7-0. *Training*. 29 April 2024.

FM 6-27. *The Commander's Handbook on the Law of Land Warfare*. 07 August 2019

FM 6-99. *U.S. Army Report and Message Formats*. 17 May 2021.

TC 3-04.11. *Commander's Aviation Training and Standardization Program*. 14 April 2022.

TP 350-70-1. *Training Development in Support of the Operational Training Domain*. 12 February 2019.

WEBSITES

Army Publishing Directorate website. <https://armypubs.army.mil/>.

Reconnaissance, Attack, Gunnery, Employment Branch SharePoint. <https://armycitaas.sharepoint-mil.us/sites/TR-ACOE-DOTD/SitePages/Gunnery-Branch.aspx>.

PRESCRIBED FORMS

This section contains no entries.

REFERENCED FORMS

Unless otherwise indicated, DA forms are available on the Army Publishing Directorate website at <https://armypubs.army.mil/>.

DA Form 2028. *Recommended Changes to Publications and Blank Forms*.

DA Form 4507. *Crew Member Grade Slip*.

DA Form 4507-1. *Maneuver/Procedure Grade Slip*.

DA Form 7122. *Crew Member Training Record*.

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TC 3-04.1-2

01 June 2026

By Order of the Secretary of the Army:

CHRISTOPHER C. LANEVE

*General, United States Army
Vice Chief of Staff*

Official:

A handwritten signature in black ink, appearing to read 'M. Sannito', with a stylized flourish at the end.

MATTHEW L. SANNITO

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to the Secretary of the Army
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